

Live Demonstration Instructions

Due: Week 11/12

Each team/group must deliver a 20-minute live demonstration of its final product using the MIEEE-13NF. You can use your power-point presentation as submitted in Week 10 ([presentation submission](#)) to complement your final demonstration.

In your demonstration, you must benchmark your results (e.g. your experiments in Module 1 and Module 3 can serve as your benchmark, as appropriate).

In your demonstration, you must provide real-time simulation results for your three completed extensions, including the mandatory (Option 3) extension to show voltage compliance at all nodes on the MIEEE-13NF.

The purpose of the demonstration is to provide convincing, technically credible evidence that the proposed deliverables have been implemented, tested, and evaluated. The client should be able to observe what the system does, understand why it behaves as it does, and see evidence that the claimed improvements have been achieved. Please refer to the [rubric](#).

Your demonstration should:

- Clearly identify the three proposed extensions and demonstrate the relevant system functionality for each one.
- Show the operation of the improved MIEEE-13NF system, including the controller, relevant measurements, battery behaviour, and grid response.
- Demonstrate that all node voltages remain within the permitted range during the required operating scenarios.
- Use suitable live evidence, such as HIL SCADA displays, controller outputs, logged signals, plots, comparison cases, or test results, to validate system performance.
- Compare the improved system with an appropriate baseline, control case, or alternative scenario to show the impact of the extensions.
- Explain the key modelling assumptions, controller design choices, implementation decisions, and engineering trade-offs.
- Describe the safety mechanisms used, including command limits, state-of-charge checks, fallback behaviour, communication-loss handling, or safe shutdown where relevant.
- Identify any limitations, incomplete features, deviations from planned deliverables, or uncertainty in the results, and explain these honestly and professionally.
- Be prepared to answer questions about the MIEEE-13NF simulation architecture, system design, validation approach, limitations, alternative approaches, and future improvements.

The demonstration must be supported by evidence. A system that is partially complete may still be demonstrated, but the team must clearly show what works, explain what does not yet work, present the available validation evidence, and demonstrate an appropriate fallback where applicable.

Teams may choose to demonstrate in Week 11 and in such cases, these teams have the opportunity to demonstrate again in Week 12. Where a team presents in both weeks, the higher mark will be used.